

DATA PREPROCESSING

Assignment 1 Report

E-commerce Customer Dataset

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Task 1. Dataset Understanding

Objective. The objective was to understand the structure and content of the customer dataset before applying preprocessing techniques.

1.1 Steps Performed

- Loaded the dataset into a Pandas DataFrame using `pd.read_csv()`.
- Displayed the first five records with `head()`.
- Checked dimensions with `shape` and listed attribute names with `columns`.
- Reviewed data types and non-null counts with `info()`.
- Generated descriptive statistics for numerical attributes using `describe()`.
- Generated categorical summaries using `describe(include='object')`.
- Separated numerical and categorical attributes with `select_dtypes()`.

Table 1. Dataset structure

Item	Result
Initial dimensions	2,200 rows x 25 columns
Content	Customer demographic, purchasing, membership, device, payment, loyalty, satisfaction, coupon, and churn attributes
Attribute names	CustomerID, CustomerName, Age, Gender, AnnualIncome_INR, IncomeCurrency, SpendingScore_1_100, MembershipTier, JoinDate, LastPurchaseDate, VisitFrequency, AvgBasketValue_INR, TotalPurchases, OnlinePurchases, StorePurchases, PreferredCategory, City, Country, EmailProvider, DeviceType, PaymentMethod, LoyaltyPoints, SatisfactionRating_1_5, CouponUsed, ChurnNextMonth
Attribute groups	Numerical, categorical/text, date, identifier, and binary attributes

1.2 Brief Explanation of the dataset

The dataset contains customer information collected by an e-commerce company and includes both numerical and categorical attributes. The preprocessing objective is to improve accuracy, consistency, and suitability for machine learning by addressing missing values, duplicate records, inconsistent categories, invalid entries, incorrect data types, and outliers. The first displayed records and descriptive statistics are retained in the accompanying notebook, while this report presents the substantive observations and decisions.

2. Data Quality Assessment

2.1 Objective

The objective of this task was to identify common data quality issues that may affect analysis and machine learning performance.

2.2 Steps Performed

The following data quality checks were carried out:

- Missing values
- Duplicate records
- Inconsistent categorical values
- Outliers using the IQR method
- Invalid entries based on domain rules
- Data type issues

The findings are summarized below

Table 2. Summary of data-quality issues

Problem	Columns affected	Count	Impact
Missing values	Age; Gender; AnnualIncome_INR; IncomeCurrency; MembershipTier; LastPurchaseDate; VisitFrequency; AvgBasketValue_INR; TotalPurchases; City; Country; EmailProvider; DeviceType; PaymentMethod; LoyaltyPoints; SatisfactionRating_1_5; CouponUsed; ChurnNextMonth	2,824	May cause incomplete or biased analysis.
Duplicate records	Entire dataset	64	May repeat information and bias results.
Inconsistent categorical values	CustomerName; Gender; MembershipTier; VisitFrequency; PreferredCategory; City; Country; EmailProvider; DeviceType; PaymentMethod; CouponUsed; IncomeCurrency; inconsistent date values	11	Equivalent categories may be treated as distinct.
Outliers	Age; AnnualIncome_INR; AvgBasketValue_INR; TotalPurchases; OnlinePurchases; StorePurchases; LoyaltyPoints; SatisfactionRating_1_5; ChurnNextMonth	1,193	Extreme values may affect statistical results and model performance.
Invalid entries	Age; AnnualIncome_INR; Gender	261	Values violate expected domain rules.
Data-type issues	Age; JoinDate; LastPurchaseDate; TotalPurchases; ChurnNextMonth; SatisfactionRating_1_5	6	Incorrect types prevent valid date, numeric, or binary operations.

Table 3. IQR outlier summary

Column	No. of outliers	Minimum outlier	Maximum outlier
Age	63	-5	180
AnnualIncome_INR	87	1,746,742	99,999,999
SpendingScore_1_100	0	-	-
AvgBasketValue_INR	139	5,995.93	999,999
TotalPurchases	111	56	9,999
OnlinePurchases	145	27	104
StorePurchases	145	30	110
LoyaltyPoints	19	2,947	999,999
SatisfactionRating_1_5	30	8	11
ChurnNextMonth*	454	1	1

2.3 Key Inconsistencies

The 11 inconsistent categorical fields included Gender, IncomeCurrency, MembershipTier, VisitFrequency, PreferredCategory, City, Country, EmailProvider, DeviceType, PaymentMethod, and CouponUsed. Problems included capitalization, spelling, spaces, abbreviations, and alternate names. Standardization converted each equivalent representation into a single accepted category.

2.4 Observations

- Missing values were identified in several columns and may require appropriate handling before model building.
- Duplicate records were checked to avoid repeated observations.

- Inconsistent categorical values (such as different capitalizations or formatting) were detected and should be standardized.
- Outliers were identified using the Interquartile Range (IQR) method. These values may influence statistical analysis and machine learning models.
- Invalid entries were identified based on predefined domain rules, such as unrealistic ages or spending scores outside the valid range.
- Date columns (JoinDate and LastPurchaseDate) are currently stored as object data types and should be converted to datetime format during preprocessing.

3. Data Cleaning

3.1 Objective

To identify and correct missing values, duplicates, inconsistent categories, invalid entries, incorrect data types, and outliers so that the dataset is accurate and reliable.

A separate `df_clean` copy preserved the source data. Table 5 records the original before/after results, reason

3.2 Observation

Table 5. Data-cleaning steps: before, after, reason, and Python code

Step	Before	After	Reason
Working copy	Operations would otherwise alter the original DataFrame.	A separate <code>df_clean</code> copy was created.	Preserve the source data and provide a controlled preprocessing workspace.
Missing-value imputation	2,824 missing values were present across numerical and categorical attributes.	Numerical gaps were filled with the median; categorical gaps with the mode. The post-imputation check returned zero missing values.	The median is robust to the extreme income and basket-value observations; the mode is appropriate for categorical variables.
Duplicate removal	The dataset contained 2,200 rows and 64 duplicate records.	Duplicates were removed; the resulting dataset contained 2,136 rows and 25 columns.	Ensure that each row represents a unique customer and avoid biased analysis.
Category standardization	Equivalent values differed in spelling, capitalization, spacing, abbreviations, and alternate names.	The affected categorical columns were converted to consistent labels, for example weekly -> Weekly, M -> Male, Silvr -> Silver, and Grocery -> Grocery.	Prevent identical categories from being treated as separate features.
Data-type conversion	TotalPurchases was stored as float64; JoinDate and LastPurchaseDate were stored as objects; ChurnNextMonth was stored as float64; Age and SatisfactionRating were also initially stored as floats.	Valid numeric values were converted to integers, and date columns were converted to datetime after invalid entries were handled.	Enable correct date calculations and numerical or binary operations without inappropriate conversion.
Noise handling	Invalid ages, negative income, invalid spending scores, and invalid ratings were present.	Invalid values were replaced with the median calculated from valid observations; affected fields were then assigned appropriate data types.	The median limits the influence of extreme values while preserving all records.
Outlier treatment	IQR analysis identified extreme values. Treatment was required for Age, AnnualIncome_INR, AvgBasketValue_INR, and LoyaltyPoints.	Values outside the lower and upper IQR limits were capped using <code>clip()</code> ; no selected values remained outside the limits.	Reduce the influence of extreme observations while retaining every record. Count-based features were not capped because extremes may reflect valid behaviour.
JoinDate final validation	<code>pd.to_datetime(..., errors="coerce")</code> converted 30 invalid values to NaT: 11 "not available", 10 instances of 31/02/2024, and 9 instances of 2024-13-01.	NaT values were replaced with the median valid date; JoinDate contained no missing values and was stored as datetime.	Preserve all records while correcting invalid date values discovered during final validation.

Category standardization was restricted to fields with repeated categorical values. CustomerID, CustomerName, JoinDate, and LastPurchaseDate were excluded from this step because they are an identifier, free-text names, and date fields respectively. Final validation confirmed that the cleaned data was consistent and ready for transformation.

3.3 Conclusion

The data-cleaning process successfully addressed missing values, duplicates, inconsistent categories, incorrect data types, invalid values, outliers and invalid dates. The final dataset contained 2,136 records and 25 attributes, with no remaining missing values, duplicate rows or invalid domain values. It was therefore suitable for the data-transformation and machine-learning preparation stages.

4. Data Transformation

4.1 Objective

To convert the cleaned data into a machine-learning-ready format through categorical encoding and numerical feature scaling.

Three transformations were applied: **Binary Label Encoding**, **One-Hot Encoding of nominal variables**, and **Min-Max Normalization** of numerical variables.

4.2 Observation

Gender was not label encoded because Male, Female, and Other are unordered nominal categories; assigning 0, 1, and 2 would imply an artificial ranking.

Table 7. Transformation comparison

Transformation	Before	After	Reason
Label encoding	CouponUsed contained Yes and No.	No = 0; Yes = 1.	A binary categorical variable can be represented numerically without implying an inappropriate order.
One-hot encoding	Nine nominal attributes contained unordered categories: Gender, IncomeCurrency, MembershipTier, VisitFrequency, PreferredCategory, City, EmailProvider, DeviceType, and PaymentMethod.	Each category became a binary column; drop_first=True reduced redundancy. The encoded dataset contained 57 columns.	Avoid arbitrary ordinal meaning and reduce multicollinearity.
Min-Max normalization	Selected numerical features had different units and ranges.	Values were scaled to the interval [0, 1] while their relative differences were preserved.	Prepare comparable inputs for distance-based analysis and PCA.

4.3 Conclusion

Label Encoding assigned No = 0 and Yes = 1 in CouponUsed. One-Hot Encoding converted the nine nominal fields to binary columns with drop_first=True, producing a final encoded dataset with 57 columns. Min-Max normalization scaled the selected numerical features to the range 0-1 while preserving their relative differences.

5. Data Reduction

5.1 Objective

To reduce the number of features and records while preserving the most important information, thereby improving computational efficiency and simplifying further analysis.

5.2 Observation

Correlation analysis was first evaluated at thresholds of 0.90 and 0.80. No highly correlated numerical features were identified, so correlation-based removal did not eliminate any columns. **PCA and Random Sampling** were therefore selected as the two reduction methods.

5.2.1 Principal Component Analysis

PCA was applied to eight Min-Max-normalized numerical features: Age, AnnualIncome_INR, SpendingScore_1_100, AvgBasketValue_INR, TotalPurchases, OnlinePurchases, StorePurchases, and LoyaltyPoints. Categorical variables were excluded because PCA operates on numerical feature variation.

Table 8. PCA before-and-after comparison

Measure	Before PCA	After PCA
Numerical dimensionality	8 normalized features	5 principal components
Variance retained	100% of the original numerical variance	Approximately 89.22%
Dominant loading by component	Original feature space	PC1: AvgBasketValue_INR; PC2: AnnualIncome_INR; PC3: SpendingScore_1_100; PC4: LoyaltyPoints; PC5: Age

5.2.2 Random Sampling

Random sampling reduced the number of records to 80% of the cleaned dataset using a fixed random state for reproducibility. The comparison was 2,136 records before sampling and an 80% subset afterwards (approximately 1,709 records). This reduced processing time while retaining a randomly selected representation of the data.

Reason for selection: PCA reduced numerical dimensionality while preserving most of the variance, and random sampling reduced the number of records without applying a deterministic preference to particular customers.

6. Proximity Measures

6.1 Objective

To measure the similarity or dissimilarity between customer records and identify customers with the most similar behavioural and purchasing profiles.

Two distance measures were calculated using **Euclidean Distance** and **Manhattan Distance** for 20 selected customers using normalized numerical attributes.

6.2 Observation

Identifiers and categorical variables were excluded because they do not represent meaningful continuous numerical distances. Smaller distances indicate greater similarity.

Customers 10 and 18 were the most similar pair under both measures. Their Euclidean distance was 0.21, indicating the smallest straight-line difference among the selected 20 customers. Although Euclidean and Manhattan metrics use different calculations, both consistently identified the same pair, strengthening the conclusion that these two customers have the most similar normalized numerical profiles.

7. Reflection Report

1. Major preprocessing challenges

The main challenges were the simultaneous presence of missing values, duplicate records, inconsistent categorical labels, incorrect data types, invalid domain values, and numerical outliers. Numerical missing values were imputed with the median, whereas categorical missing values were imputed with the mode. This choice was suitable because several numerical variables contained extreme values that could distort the mean. Duplicate rows were removed to prevent repeated observations from influencing the analysis. Category standardization was particularly demanding because the same category appeared with different capitalization, spacing, spelling, abbreviations, and alternate names. Date fields also required careful conversion because invalid strings became missing datetime values. Finally, outliers had to be distinguished from potentially valid customer behaviour before treatment.

2. Preprocessing step with the greatest impact

Data cleaning had the greatest overall impact because every later transformation depended on the quality of the cleaned data. Imputation, duplicate removal, category standardization, type conversion, invalid-value correction, and outlier treatment produced a consistent dataset that could be transformed reliably. Label Encoding and One-Hot Encoding then converted categorical information into machine-readable form, while Min-Max normalization made numerical features comparable. PCA reduced eight normalized numerical features to five

principal components and retained approximately 89.22% of the original variance. These operations made the dataset more suitable for machine-learning methods without changing the substantive findings of the analysis.

3. Problems remaining after preprocessing

Some limitations remained after preprocessing. Median and mode imputation are practical, but the imputed values may not reproduce the unknown original observations. IQR-based capping may also modify legitimate extreme customer behaviour because unusual values are not always errors. PCA preserved most of the information, but reducing eight features to five components necessarily lost a small amount of variance. In addition, the resulting principal components are less directly interpretable than the original variables. Therefore, the dataset became substantially cleaner and more consistent, but some information loss and modelling assumptions remain unavoidable.

4. Recommended machine-learning algorithm

K-Means Clustering is recommended for the next stage. The dataset contains customer characteristics such as annual income, spending score, purchase behaviour, basket value, and loyalty points, which are appropriate for customer segmentation. The numerical features were normalized and reduced using PCA, so the data is prepared for a distance-based algorithm. The proximity analysis also demonstrated that customer similarity can be measured consistently: both Euclidean and Manhattan distance identified Customers 10 and 18 as the most similar pair. These results support the use of clustering to identify customer groups with similar behavioural profiles.

5. Additional preprocessing before deployment

Before deployment, the data should be checked for class imbalance, feature importance, and new inconsistencies that may appear in incoming records. The entire workflow should be implemented as an automated preprocessing pipeline so that training and future prediction data receive exactly the same transformations. Cross-validation should be used during model evaluation, and performance should be monitored on unseen data to identify drift or deterioration. These additional controls would improve the reliability, reproducibility, and robustness of the final machine-learning system.